RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 2 (Delta)
Topic(s): Complex Numbers and Sequences and Series (Questions)

1 The sum, S_n, of the first n terms of a sequence $u_1, u_2, u_3, \ldots$ is given by $S_n = n(2n+c)$, where c is a constant.

(i) Find u_n in terms of c and n. [3]

(ii) Find a recurrence relation of the form $u_{n+1} = \mathrm{f}(u_n)$. [2]

2 A sequence $u_1, u_2, u_3, \ldots$ is given by

$$u_1 = 2 \text{ and } u_{n+1} = \frac{3u_n - 1}{6} \text{ for } n \geq 1.$$

(i) Find the exact values of u_2 and u_3, and the value of u_{10} to 5 decimal places. [3]

(ii) It is given that $u_n \to l$ as $n \to \infty$. Showing your working, find the exact value of l. [2]

(iii) Describe the behaviour of the sequence. [1]

3 A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n(1 - u_n)$$

for all positive integers n.

(a) Find the possible limits of the sequence, if the sequence is convergent. [2]

(b) For each of the following, evaluate u_2 and u_3, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

4 A sequence of positive numbers $x_1, x_2, x_3, \ldots, x_n$ is a geometric progression. Show that the sequence of numbers $y_1, y_2, y_3, \ldots, y_n$ given by the relation,

$$y_n = \log_k x_n + k, \quad \text{where } k \text{ is a positive constant,}$$

is an arithmetic progression. [4]

5 **(a)** **(i)** Find $\sum_{r=1}^{k}\left[\left(-\frac{1}{2}\right)^{r+1} + \ln(r+1)\right]$ in terms of k. Simplify your answer. [4]

(ii) Hence determine if $\sum_{r=1}^{\infty}\left[\left(-\frac{1}{2}\right)^{r+1} + \ln(r+1)\right]$ exists. [1]

(b) The first term of an arithmetic series is positive. The sum of the first 6 terms of the series is 4.5, and the product of the first four terms of the series is 0. Find the 13th term of the series. [4]

6 A sequence of numbers $u_1, u_2, u_3, \ldots$ has a sum S_n where $S_n = \sum_{r=1}^{n} u_r$.

(i) It is known that the sum of the first three terms of the sequence is 17 and $\sum_{n=1}^{3} S_n = 30$.

Furthermore, the third term of the sequence is twice the first term of the sequence. Find the first three terms of the sequence. [3]

(ii) A constant k is subtracted from each term of the sequence so that the first three terms of the modified sequence are now three consecutive terms of a geometric progression with a common ratio greater than 1.

Find the value of k and the common ratio of the geometric progression. [3]

7 An arithmetic sequence has first term a and common difference d, where a and d are non-zero. The ninth, tenth and thirteenth terms of the arithmetic sequence are the first three terms of a geometric sequence.

(i) Show that $a = -\frac{15}{2}d$. [3]

(ii) The sum of the first n terms of the arithmetic sequence is denoted by S_n. Find the value of S_{16}. [2]

(iii) Given that the kth term of the arithmetic sequence is the fourth term of the geometric sequence, find the value of k. [3]

8 Mr Chan is interested in investing in a savings account with an interest rate of 3.2% per year, so that on the last day of each year, the amount in the savings account on that day is increased by 3.2%. He decided to invest \$8 000 on the first day of each year, starting from 2020.

(i) Show that the amount in the account at the end of n years after the interest has been added is given by $\$\,258\,000(1.032^n - 1)$. [3]

After 10 years, Mr Chan stopped investing in this savings account. If he does not withdraw any money from this savings account, the savings account will still continue to generate interest of 1.5% per year on the amount in the account, so that on the last day of each year the amount in the account on that day is increased by 1.5%. Assuming that Mr Chan does not withdraw any money from the savings account,

(ii) by the end of which year will the total interest be first more than \$35 000 from the day that Mr Chan first started his saving account? [3]

At the age of 55, Mr Chan is able to receive a monthly pay-out over a period of 20 years from the savings account if he did not withdraw before. The monthly pay-out in the first year is \$850. The monthly pay-out for each subsequent year is an increment of \$$D$ from the monthly pay-out of the previous year. The total pay-out to Mr Chan at the end of 20 complete years is \$352 200.

(iii) Given that the monthly pay-out is \$1 500 for the m th year of the 20 year period, where m is a positive integer, find m. [3]

9 Mary and Leo signed up for a 42 km marathon and decided to run every day to train for the marathon.

On the first day of Mary's preparation, she begins her training by running 12 km. For each odd numbered day of training, she increases the distance by 2km each time. For each even numbered day of training, she runs the same distance as the day before. Her daily distance ran is then $\{12,12,14,14,16,16,\cdots\}$.

(i) Show that Mary first runs 42 km on the 31st day of training. [2]

Leo starts his training with a 12 km run on the same day as Mary, but decides to increase the distance every day by $x\%$ such that he would be running 42 km on the same day as Mary first runs 42 km.

(ii) Find the value of x, giving your answer correct to 2 decimal places. [2]

(iii) Find the total distance covered by both runners on the day they have completed their first 42 km run at the end of the training. [3]

10 **Do not use a calculator in answering this question.**

The complex number z satisfies the equation

$$z^2-(4+\mathrm{i})z+2(\mathrm{i}-t)=0,$$

where t is a real number. It is given that one root is of the form $k-k\mathrm{i}$, where k is real and positive. Find t and k, and the other root of the equation. [7]

11 Two complex numbers w and u satisfy the equations

$$w^*-2\mathrm{i}u=8 \quad \text{and} \quad (2\mathrm{i}-1)w+2u^*=4.$$

Find w and u, giving your answers in the form $x+\mathrm{i}y$, where x and y are real. [4]
Hence find $|u^*-w|$ and $\arg(u^*-w)$. [2]

12 Given $u=1+\sqrt{3}\mathrm{i}$ and $v=1-\mathrm{i}$ are roots of the equation $z^4+az^3+bz^2+cz+d=0$, where a, b, c and $d \in \mathbb{R}$,

(a) find the values of a, b, c and d. [4]

The complex numbers $\mathrm{i}u$ and v^* are represented by the points P and Q respectively on an Argand diagram.

(b) Find the area of triangle OPQ. [3]

13 The points P and Q are represented by the complex numbers p and q respectively, where

$$p = \sqrt{2} + \sqrt{2}\mathrm{i}\ ,\ \arg(q) = \frac{2}{3}\pi \text{ and } |q| = 2\,.$$

(a) Find $|p|$ and $\arg(p)$ in exact form. [2]

(b) Sketch the points P and Q on an Argand diagram. [2]

(c) Use the Argand diagram, or otherwise, to find $\mathrm{Re}(q)$ and $\mathrm{Im}(q)$, giving your answers in exact form. [2]

(d) The point R is represented by the complex number $p+q$. What can you deduce about the shape of quadrilateral $OPRQ$, where O is the origin? [1]

(e) By considering $\arg(p+q)$ or otherwise, show that $\tan\left(\frac{11}{24}\pi\right) = \sqrt{6}+\sqrt{3}+\sqrt{2}+2$. [3]

14 **Do not use a calculator in answering this question.**

It is given that $z = \sqrt{3} - \mathrm{i}$ and $w = 2\mathrm{i}z$.

(a) Find the modulus and argument of z. [2]

O, A, B and C represent the complex numbers 0, z, $z+w$ and w respectively.

(b) Illustrate these four points on a single diagram. [2]

(c) Deduce the modulus and argument of w. [2]

(d) Identify the shape of the quadrilateral $OABC$ and state its area. [2]

15 **Do not use a calculator in answering this question.**

(i) Find the values of z and w that satisfy the equations $(1+\mathrm{i})z + 2w = -2 + 4\mathrm{i}$ and $3z - w = 4 + 2\mathrm{i}$, expressing your answers in the form $c + d\mathrm{i}$, where $c, d \in \mathbb{R}$. [4]

(ii) Points W and Z represent w and z found in part **(i)**. Find $\frac{w}{z}$ in the form $p+\mathrm{i}q$, where $p, q \in \mathbb{R}$. Hence, state the transformation that maps line segment OZ onto line segment OW. [2]